

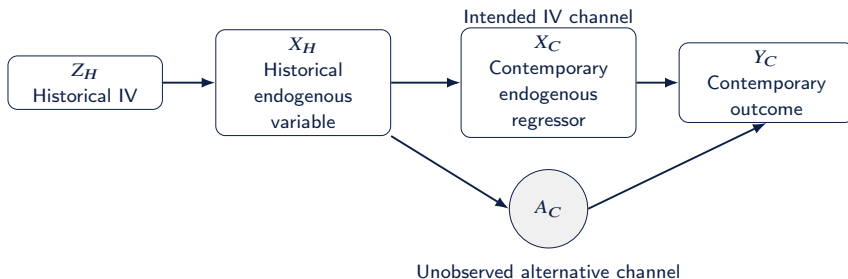
Historical Instruments and Exclusion Restrictions: A Replication Study of Roche (2020)

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Why historical instruments can be problematic?

- Casey and Klemp (2021) study IV regressions where historical instruments are used for contemporary endogenous regressors
- Their main point is that the exclusion restriction can fail because of the time gap between the instrument's initial impact and the measurement of the endogenous regressor



- Ask how sensitive Roche (2020)'s IV estimates are to violations of the exclusion restriction
- Re-examine Roche (2020) allowing for imperfect instrument validity
- Estimate the range of treatment effects under partial exclusion (Falsification Adaptive Set)
- Compare the conventional 2SLS estimate with the FAS interval to assess robustness

Part I: Theoretical Framework

- Falsifiable models: There exists a population distribution of the observable data which is inconsistent with the model
- What should researchers do when their baseline model is falsified?
- This paper provides a way to salvage a falsified baseline model by considering continuous relaxations of the baseline assumptions
- By sufficiently weakening the assumptions, a falsified baseline model becomes nonfalsified

- Suppose the model has L assumptions
- For each assumption $l \in \{1, 2, \dots, L\}$, define a class of assumptions indexed by δ_l ($\delta_l = 0$ means that assumption l is imposed)
- Using the definition above, $\delta'_l \geq \delta''_l$ means that δ'_l is weaker than δ''_l
- Let $\delta = (\delta_1, \dots, \delta_L)$ and $\delta = 0$ mean the baseline model

The falsification frontier is the set

$$FF \equiv \{\delta \in \mathcal{D} : \delta \in \mathcal{D}_{nf} \text{ and for any } \delta' < \delta, \delta' \in \mathcal{D}_f\}$$

- where $\mathcal{D}_f$ is set of all assumptions which are falsified
- $\mathcal{D}_{nf}$ is set of all assumptions which are non-falsified
- For any $\delta_0 \in FF$, the model with assumption δ is falsified if $\delta < \delta_0$
- The frontier represents **the smallest relaxation necessary to reconcile the model with the data**

- Let $\Theta_I(\delta)$ denote the identified set for a parameter of interest, given the model with assumption δ
- Then we can define falsification adaptive set(FAS) by

$$\bigcup_{\delta \in FF} \Theta_I(\delta)$$

- FAS summarizes every parameter value supported by minimally relaxed assumptions

Linear Models with Multiple IVs

Consider the following linear model:

$$Y(x, z) = x'\beta + z'\gamma + U$$

- where $X \in \mathbb{R}^K$ and $Z \in \mathbb{R}^L$
- Assumption 1: $\text{Rank}(\text{Cov}(Z, X)) = K$
- Assumption 2: $\text{Var}(Z)$ is invertible
- Assumption 3: $\text{Cov}(Z_l, U) = 0 \quad \forall l = 1, 2, \dots, L$
- Assumption 4: $\gamma = 0$
- Assumption 4' (Partial Exclusion): There are known constants $\delta_l \geq 0$ such that $|\gamma_l| \leq \delta_l \quad \forall l = 1, 2, \dots, L$

Theorem

(Identified Set for β)

$$\Theta_I(\delta) = \{b \in \mathbb{R}^K : -\delta \leq \text{Var}(Z)^{-1}(\text{Cov}(Z, Y) - \text{Cov}(Z, X)b) \leq \delta\}$$

- Let $\psi = E[ZZ']^{-1}E[ZY]$ and $\pi = E[ZZ']^{-1}E[ZX]$
- The identified set when $K=1$ can be rewritten in terms of the components of ψ and π (cf. Corollary 1)

Proposition

(Falsification Frontier) Under Assumption 1-3,

$$FF = \{\delta \in \mathbb{R}_+^L : \delta_l = |\psi_l - b\pi_l|, l = 1, 2, \dots, L, b \in [\min_{l:\pi_l \neq 0} \frac{\psi_l}{\pi_l}, \max_{l:\pi_l \neq 0} \frac{\psi_l}{\pi_l}]\}$$

Theorem

(Falsification Adaptive Set) Under Assumption 1-3,

$$FAS = [\min_{l:\pi_l \neq 0} \frac{\psi_l}{\pi_l}, \max_{l:\pi_l \neq 0} \frac{\psi_l}{\pi_l}]$$

- Lemma 1 shows that $\frac{\psi_l}{\pi_l}$ is equivalent to 2SLS coefficient on X using Z_l as the excluded IV and Z_{-l} as controls

Part II: Empirical Applications

- Research Question: Does neighborhood street connectivity increase innovation?
- Roche (2020) studies whether street-network density increases innovation by making interpersonal interaction more likely and more efficient
- More connected neighborhoods produce more patents (innovative)
- Use historical instruments: The percentage of housing units built before 1940(HUp_{pre1940}) and between 1940 and 1949(HU_{1940s})

Falsification Adaptive Set (FAS)

Panel A. Estimates	
Baseline 2SLS	0.0426** (0.0188)
HUpred1940 as IV, control HU1940s	0.0504*** (0.0193)
HU1940s as IV, control HUpred1940	-0.0096 (0.0449)
Panel B. Falsification adaptive set	
Relevant instruments	HUpred1940, HU1940s
First-stage F statistics	55.32, 14.78
FAS	$[-0.0096, 0.0504]$

Falsification Frontier

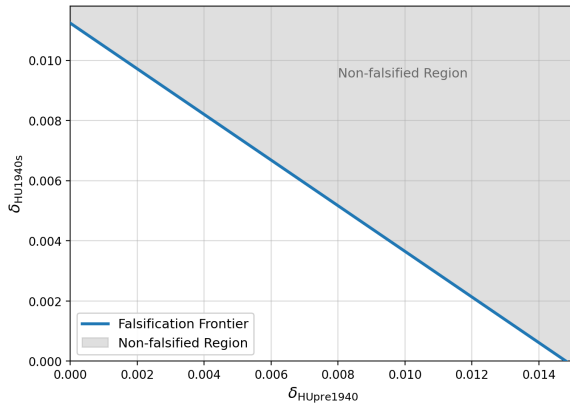


Figure 1: Falsification Frontier

- The negative endpoint is obtained when **HU1940s is used as the excluded instrument** and HUpre1940 is included as a control
- This estimate is statistically insignificant and is based on a substantially weaker first stage than the alternative controlled-IV specification
- By contrast, using **HUpre1940 as the excluded instrument** yields a positive and statistically significant estimate
- Thus, the FAS does not establish that the effect must be positive, but the evidence against Roche's positive-effect conclusion is driven by the less precisely estimated controlled-IV specification